Muzaffar Qureshi

Curriculum Vitae

Dept. of Mechanical & Aerospace Engineering
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Professional Summary

Robotics and Control researcher with expertise in Autonomous Systems, Nonlinear and Safety-Critical Control, Reinforcement Learning (RL), Gaussian Processes (GP), and UAV (Unmanned Aerial Vehicle) flight control. Experience in GPS-denied navigation, motion/trajectory planning, state estimation, and experimental validation. Proficient in MATLAB, Simulink, Python

Education

2022–2026 (In progress) **Ph.D., Mechanical Engineering**, *University of Florida*, Gainesville, FL.
Advisor: Assoc. Prof. Dr. Rushikesh Kamalapurkar; Research: Robotics, Nonlinear Control, Gaussian Processes, Machine Learning (safe RL, uncertainty quantification).

2017–2019 **M.S., Aerospace Engineering**, *Air University*, Islamabad, Pakistan.
Advisor: Prof. Dr. Suhail Akhtar; Research: Reinforcement Learning (RL), Safe Control, fault tolerance; GPA: 4.00/4.00, Rank: 1/24 (Rector's Gold Medal).

2009–2013 **B.E., Aerospace Engineering**, *National University of Sciences and Technology (NUST)*, Islamabad, Pakistan.
Advisor: Prof. Dr. Farooq Akram; Research: Design and Fabrication of a UAV (Unmanned Aerial Vehicle); GPA: 3.50/4.00, Rank: 2/44 (Rector's Silver Medal).

Publications

Journal Articles

2026 **Muzaffar Qureshi** H., Tochukwu E. Ogri, Trivikram Satharasi, Kyle Volle, and Rushikesh Kamalapurkar. Gaussian process regression under deterministic input bias. *Accepted at IEEE Robot. Autom. Lett.*, Sep 2026.

2026 **Muzaffar Qureshi**, Tochukwu E. Ogri, Wanjiku A. Makumi, Kyle Volle, and Rushikesh Kamalapurkar. A Hough transform approach to safety-aware scalar field mapping using Gaussian processes. *Under review at IEEE Transactions on Control Systems Technology (TCST)*, May 2026.

2026 **Muzaffar Qureshi**, Tochukwu E. Ogri, Wanjiku A. Makumi, Kyle Volle, and Rushikesh Kamalapurkar. GP-SCB: Confidence-based safe exploration in Gaussian process bandit optimization. *Under review for IEEE Transactions on Automation Science and Engineering (T-ASE)*, Sep 2026.

2025 **Muzaffar Qureshi**, Tochukwu E. Ogri, Humberto Ramos, Wanjiku A. Makumi, Zachary I. Bell, and Rushikesh Kamalapurkar. A switched adaptive control approach to reduce sensing needs in trajectory tracking problems. *IEEE Control Systems Letters*, volume 9, pages 2777–2782, 2025.

2025 **Muzaffar Qureshi**., A. Maqsood, and A. Fayyaz ud Din. Fault-tolerant control of a quadcopter using reinforcement learning. *SAE International Journal of Aerospace*, volume 18, pages 75–89, 2025.

2025 Tochukwu Elijah Ogri, **Muzaffar Qureshi**, Zachary I. Bell, and Rushikesh Kamalapurkar. State and input constrained output-feedback adaptive optimal control of affine nonlinear systems. *IEEE Transactions on Automatic Control*, 2025. Under review.

2025 Tochukwu E. Ogri, **Muzaffar Qureshi**, Zachary I. Bell, Wanjiku A. Makumi, and Rushikesh Kamalapurkar. An output feedback approach to differential graphical games in linear multiagent systems. *IEEE Control Systems Letters*, volume 9, pages 2265–2270, 2025.

2025 Tochukwu E. Ogri, Muzaffar Qureshi, Zachary I. Bell, and Rushikesh Kamalapurkar. Safe output-feedback adaptive optimal control of input constrained affine nonlinear systems. *Accepted for publication at Automatica*, 2025. see arXiv:2406.18804.

2023 Adnan Fayyaz Ud Din, Suhail Akhtar, Adnan Maqsood, **Muzaffar Qureshi**, and Imran Mir. Modified model free dynamic programming: An augmented approach for unmanned aerial vehicle. *Applied Intelligence*, volume 53, pages 3048–3068, 2023.

2023 Zaheer Abbas, Mohtashim Mansoor, **Muzaffar Qureshi**, and Zahid Mehmood. Review: Mems sensors for flow separation detection. *Microsystem Technologies*, volume 29, pages 1253–1280, 2023.

2022 Taimur Ali Shams, Syed Irtiza Ali Shah, **Muzaffar Qureshi**, Mohtashim Mansoor, et al. A novel solution methodology for longitudinal flight characterization of a flying-wing micro aerial vehicle. *Proc. Inst. Mech. Eng. G J. Aerosp. Eng.*, volume 236, 2022.

In Conference Proceedings

2027 **Muzaffar Qureshi**, Yihui Mao, James W. Cross, Kyle Volle, and Rushikesh Kamalapurkar. Exposure-aware informative path planning using deep reinforcement learning, 2027. Under review for the IEEE International Conference on Robotics and Automation (ICRA 2027).

2026 **Muzaffar Qureshi**, Tochukwu Elijah Ogri, Zachary I. Bell, Wanjiku A. Makumi, and Rushikesh Kamalapurkar. A Taylor series approach to correction of input errors in Gaussian process regression, 2026. Accepted for publication at European Control Conference (ECC 2026).

2026 Trivikram Satharasi, Tochukwu E. Ogri, **Muzaffar Qureshi**, Kyle Volle, and Rushikesh Kamalapurkar. Safe adaptive feedback control via barrier states. In *Proc. IFAC World Congr.*, 2026.

2026 Trivikram Satharasi, Tochukwu E. Ogri, **Muzaffar Qureshi**, Kyle Volle, and Rushikesh Kamalapurkar. Adaptive control with sparse identification of nonlinear dynamics. arXiv:2604.06338, 2026. Accepted for publication at CDC 2026.

2026 Hossein Papi, **Muzaffar Qureshi**, Kyle Volle, and Rushikesh Kamalapurkar. Decentralized scalar field mapping using gaussian process. arXiv:2604.06554, 2026. Preprint.

2025 **Muzaffar Qureshi**, Tochukwu Elijah Ogri, Humberto Ramos, Zachary I. Bell, and Rushikesh Kamalapurkar. Gaussian process-based scalar field estimation in GPS-denied environments, 2025.

2025 Tochukwu Elijah Ogri, **Muzaffar Qureshi**, Zachary I. Bell, Matthew Longmire, and Rushikesh Kamalapurkar. Monocular inspection of spacecraft under illumination constraints and avoidance regions. In *2025 American Control Conference (ACC)*, pages 1713–1718, 2025.

2025 Tochukwu E. Ogri, **Muzaffar Qureshi**, Zachary I. Bell, Wanjiku A. Makumi, and Rushikesh Kamalapurkar. Safe output-feedback adaptive optimal control of affine nonlinear systems, 2025.

2024 **Muzaffar Qureshi**, Tochukwu Elijah Ogri, Zachary I. Bell, and Rushikesh Kamalapurkar. Scalar field mapping with adaptive high-intensity region avoidance. In *2024 IEEE Conference on Control Technology and Applications (CCTA)*, pages 388–393, 2024.

2024 Tochukwu Elijah Ogri, **Muzaffar Qureshi**, Zachary I. Bell, Kristy Waters, and Rushikesh Kamalapurkar. An adaptive optimal control approach to monocular depth observability maximization. In *2024 American Control Conference (ACC)*, pages 2356–2361, 2024.

2021 Abdul Rahim Tajammal and **Muzaffar Qureshi**. Autonomous control of a quadcopter using machine learning algorithm. In *2021 Seventh International Conference on Aerospace Science and Engineering (ICASE)*, pages 1–8, 2021.

2020 Adnan Fayyaz Ud Din, Suhail Akhtar Janjua, Adnan Maqsood, and **Muzaffar Qureshi**. Design of a reinforcement learning based controller for gliding control of an experimental design vehicle. In *AIAA Scitech 2020 Forum*, Orlando, FL, United States, January 2020. American Institute of Aeronautics and Astronautics (AIAA). AIAA Paper 2020-1849.

Research Experience

Jun 2026 – Aug 2026 **Robotics Research Intern — AFRL Scholarship**, *REEF / Air Force Research Laboratory (AFRL)*, Shalimar, FL.
Developing a reinforcement learning-based safe and informative sampling framework for autonomous systems operating in uncertain environments. Implementing and evaluating Proximal Policy Optimization (PPO) and Advantage Actor-Critic (A2C) algorithms to learn sampling policies that balance safety constraints with information acquisition; manuscript in preparation for submission to ICRA 2027.

Jun 2024 – Aug 2024 **Robotics Research Intern — AFRL Scholarship**, *REEF / Air Force Research Laboratory (AFRL)*, Shalimar, FL.
Developed a switched adaptive observer and control framework for trajectory tracking under GPS-available and GPS-denied operating modes, using intermittently acquired state measurements to reduce sensing requirements while maintaining tracking performance. This work resulted in the publication "A Switched Adaptive Control Approach to Reduce Sensing Needs in Trajectory Tracking Problems," *IEEE Control Systems Letters*, vol. 9, pp. 2777–2782, 2025.

Jul 2016 – Aug 2017 **Certification Engineer**, *Airworthiness Certification Authority*, Pakistan.
Technical Reviewer for aircraft modification/integration; ground and flight tests verification and airworthiness certification procedures; safety/compliance, fault detection and isolation (FDI), stability and control assessments.

Jan 2014 – Jun 2016 **Aircraft Maintenance Engineer**, *Engineering Wing*, Pakistan.
Supervised aircraft technical trades; troubleshooting engine/control defects; defect trend analysis; emergency handling of aircraft/UAVs; reliability, maintenance, and airworthiness support.

Teaching Experience

Sep 2019 – Aug 2022 **Lecturer**, *College of Aeronautical Engineering, NUST*, Pakistan.
Taught Heat Transfer, Aerodynamics, Flight Stability, CFD, Aerospace Engineering; supervised student research in ML, flight control, 6-DoF modeling, and CFD.

Assistantship

Fall, 2023 : **Teaching Assistant**, MAE 5433: Robotics, Kinematics, Dynamics and Control, Oklahoma State University.

Projects

Jun 2026 **Safe and Information Path Planning for Autonomous Robots**, *AFRL Project*.
Safe and budget-constrained estimation of scalar fields using autonomous robots with limited communication topology; real-time sensing and motion planning with GP and experimental validation.

Jul 2025 **Safe Control of Ground Robots in Unsafe Scalar Fields**, *AFRL Project*.
Safety-constrained control for autonomous ground robots in hazardous scalar fields; real-time sensing, motion planning, trajectory optimization; uncertainty-aware planning with GP; experimental validation.

Jul 2021 **Gliding Control of Experimental Vehicle using RL**, *AIAA SciTech Forum Project*.
Reinforcement Learning (actor–critic/policy gradient) to maximize glide performance; nonlinear simulation, reward design, model-free control; robustness analysis.

Aug 2019 **Stability and Control of Quadcopter under Propeller Failure**, *M.S. Thesis, Air University*.
Fault-tolerant control under actuator failure; stability analysis (Lyapunov), observer/estimation, robust/adaptive control; simulation and experimental validation.

May 2013 **Design and Fabrication of a Fire Fighting UAV**, *Future Flight Design Competition, Ankara, Turkey*.
Autonomous fire detection/suppression UAV; sensing, guidance, and control integration; successful prototype demonstration.

Technical Skills

Programming	Python, MATLAB, Simulink, Julia
Control	Nonlinear Control, MPC, CBF, Adaptive/Robust Control, Kalman/EKF/UKF, State Estimation
ML/GP	Reinforcement Learning (RL), Actor–Critic, Policy Gradient, Q-learning, Gaussian Processes (GP), Bayesian Regression, Uncertainty Quantification
Data/Tools	Microsoft Office, Visio, OriginPro

Certifications

Matrix Algebra for Engineers — The Hong Kong University of Science and Technology, Dec 2025, Credential ID: 9AT48N2MB4I1

Introduction to Mathematical Thinking — Stanford University, Nov 2025, Credential ID: KCSTDHF5HTMH

Introduction to Machine Learining — Duke University, Mar 2026, Credential ID: XTN2M5T2K20B

Physical Science Responsible Conduct of Research Course 1 — CITI Program, Oct 2022, Credential ID: 52291052

Undue Foreign Influence: Risks and Mitigations — CITI Program, Oct 2022, Credential ID: 52291053

Academic Integrity for International Students — Oklahoma State University, Sep 2022

Memberships

Pakistan Engineering Council (PEC)-Professional Engineer (Aeronautical Engineering), Reg. No: 1728

IEEE-Student Member, Membership No: 99582698

IEEE Young Professionals — Certificate of Membership, 2024

IEEE Control Systems Society — Graduate Student Member, 2024

Profiles

Google Scholar

LinkedIn

ORCID

Fellowships & Awards

2025 Recipient of the **Sung and Yvonne Lu Fellowship**, Department of Mechanical and Aerospace Engineering, University of Florida.

Referees

Dr. Rushikesh Kamalapurkar
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